

Why we collect these statistics

PURPOSE OF VACUUM

- Clean dead versions of tuples
- Update the visibility map (VM)
- Update the free space map (FSM)
- Freeze tuples, blocks and tables
- Update catalog information

MAIN DIFFICULTIES

- Long-running transactions**
hold back the xmin horizon - vacuum can't reclaim recently dead tuples
- Bloated indexes**
every vacuum re-reads the entire index
- Many parameters**
hard to predict the effect of chaching them without statistics

THE (AUTO)VACUUM OBSERVABILITY GAP

pg_stat_all_tables

- Counts
 - vacuum_count, n_dead_tup, last_vacuum
- No I/O
- No WAL

pg_stat_progress_vacuum

- Current phase
- In-flight only
- No history

Server log

- Has the numbers
- Unstructured text
- Memory consuming

What statistics we add

COMMON Shared by all three views

- WAL** wal_records · wal_fpi · wal_bytes
- BUFFER I/O** blks_fetched · blks_hit · blks_read
blks_dirtied · blks_written
- TIME** total_time · delay_time
- COUNTERS** interrupts_count wraparound_failsafe_count

pg_stats_vacuum_database

- ALL COMMON FIELDS (TABLES + INDEXES)**
vacuum_count · per-DB failsafe
interrupts · delay_time

pg_stats_vacuum_tables

- ALL COMMON FIELDS**
 - PAGES** pages_scanned · pages_removed
 - VISIBILITY MAP** vm_new_visible · vm_new_frozen
 - TUPLES** tuples_deleted · tuples_frozen
recently_dead · missed_dead_tuples

pg_stats_vacuum_indexes

- ALL COMMON FIELDS**
tuples_deleted · pages_deleted



SCAN · PATCH #5012

How it lands in core

Vacuum stages · Gather vacuum statistics

DBA · READ & ACT

FILTERS & ANOMALIES HUNTING

DBA CONTROLS

- track_relations = tables | indexes | both relation_whitelist = orders, events
- track_counters = wal, timing, buffers, ... database_whitelist = mydb, analytics



I see a **WAL spike** on **orders** at 03:17 — and zero blocks read. Probably **maintenance_work_mem** too small. Let's check

wal_bytes per vacuum cycle MB

- 03:17 spike

delay_time / total_time % %

- 95% throttle!

blks_read per cycle pages

- blks_read=2

TRACKING DYNAMICS & ANOMALIES ARE POSSIBLE NOW...

Total time vs Delay

Visibility Map Marks

1 Load extension

- Add custom vacuum statistics EXT_VAC_DB and EXT_VAC_REL

2 Filters

- check if stats should be saved

3 Hook calls the function that writes to hash table

- The hook receives the diff, applies the filters from ①, then takes the per-entry LWLock and adds counters into the dhash.

4 Cleans heap & indexes, calls report() via hook

- Each phase ends with a snapshot diff of statistics or new accumulated counters.

5 SQL views

- pg_stats_vacuum_tables
- pg_stats_vacuum_indexes
- pg_stats_vacuum_database

4 Flush to disk

- Checkpoint & shutdown serialize the dhash atomically.

#	KIND	DATABASE	OBJECT	COUNTERS (CUMULATIVE)	
1	RELATION	mydb	orders	seq_scan: 42	tup_ins: 15203
2	FUNCTION	mydb	calc_total	calls: 5840	total_time: 12.4s
3	DATABASE	mydb	—	xact_commit: 48201	blks_read: 9104
4	EXT_VAC_REL	mydb	orders	blks_read: 340	pages_scanned: 128
5	EXT_VAC_REL	mydb	orders_pkey	blks_read: 52	pages_deleted: 3
6	EXT_VAC_DB	mydb	—	blks_read: 392	wal_bytes: 81920
...	...	...	...	...	...

Time spent by vacuum · Declining trend

Cache block reads

Cache hits